

Daheng Yin

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Education

Simon Fraser University

Ph.D. in Computing Science
Advisor: Prof. Jiangchuan Liu
2023.08 -- Present

Southeast University

M.Eng. in Computer Science
Advisor: Prof. Fang Dong
2020.08 -- 2023.07

Jiangnan University

B.Eng. in IoT Engineering
Advisor: Prof. Ya Guo
2016.08 -- 2020.07
GPA: 3.63, Rank 9/141

Research Areas

- Volumetric Video Streaming
- Dynamic 3D Reconstruction
- Embodied AI Systems

Links

Homepage:
yindaheng98.top
Github: [@yindaheng98](https://github.com/yindaheng98)

Research Profile

- I build real-time 4D spatial perception infrastructure for Physical AI: systems that re-construct, compress, and stream photorealistic 4D spatial representations of dynamic physical environments, enabling Physical AI systems to perceive the physical world, interact with it, and access it remotely.

Selected Publications

CAGS: Color-Adaptive Volumetric Video Streaming with Dynamic 3D Gaussian Splatting SIGGRAPH, 2026

- **Daheng Yin**, Yili Jin, Jianxin Shi, Isaac Ding, Miao Zhang, Fangxin Wang, Zhaowu Huang, Cong Zhang, Jiangchuan Liu, Fang Dong
- We propose color-adaptive 3DGS streaming for remote physical environments: compresses dynamic 3DGS volumetric video into vector-quantized LoD representations and restores color fidelity on the client, preserving photorealistic access under fluctuating bandwidth.

TrackerSplat: Exploiting Point Tracking for Fast and Robust Dynamic 3D Gaussians Reconstruction SIGGRAPH Asia, 2025

- **Daheng Yin**, Isaac Ding, Yili Jin, Jianxin Shi, Jiangchuan Liu
- High-performance dynamic 3D scene reconstruction for fast-moving scenes with point tracking: lifts cross-frame point trajectories onto 3D Gaussians and guides their translation, rotation, and scale compensation, reducing drift, artifacts, and temporal flickering.

FSVFG: Towards Immersive Full-Scene Volumetric Video Streaming with Adaptive Feature Grid ACM Multimedia, 2024

- **Daheng Yin**, Jianxin Shi, Miao Zhang, Zhaowu Huang, Jiangchuan Liu, Fang Dong
- Adaptive feature-grid streaming for large-scale full-scene volumetric video: represents wide-area dynamic scenes with compact feature grids and selectively transmits features and residuals, balancing scene coverage, visual quality, and transmission cost.

Teaching Experiences

CMPT 310 Introduction to Artificial Intelligence Summer 2026

- Topics: Search Algorithms, Supervised Learning, Gradient Descent, Markov Decision Processes, Bayesian Networks.

CMPT 365 Multimedia Systems Fall 2024

- Topics: Multimedia Representation, Audio/Video Compression, Image and Video Processing, Multimedia Communication.

CMPT 300 Operating Systems I Fall 2023

- Topics: Process Management, Synchronization, Memory Management, File Systems, Multiprogramming.

Contests & Honors

2025	SIGGRAPH Asia Volumetric Video Workshop	Compression Track 3rd
2022	TensorRT Hackathon (NVIDIA & TIANCHI)	Winner Prize
2021	TensorRT Hackathon (NVIDIA & TIANCHI)	Rank 4/48
2020	Outstanding Graduate, Jiangnan University	
2018	National College Mathematical Contest in Modeling	2nd Prize (National)
2017	China National Scholarship (2016-2017)	
2017	9th National College Mathematical Contest	2nd Prize (Provincial)
2017	14th Jiangsu College Mathematical Contest	1st Prize